

SUMMATIVE ASSIGNMENT

Econ 44815_2026
Advanced Macroeconomics, 2026
Summative Assignment (Main)
Word Limit: 1000 words

Consider the following social planning problem:

$$\max E_t \sum_{s=0}^{\infty} \beta^s \left[\frac{Z_{t+s}^c C_{t+s}^{1-\sigma}}{1-\sigma} \right]$$

subject to:

$$\text{Resource Constraint: } C_t + I_t = A_t K_t^\alpha,$$

$$\text{Investment Technology: } K_{t+1} = (1 - \delta)K_t + Z_t^x I_t.$$

where C_t = consumption at date t , I_t = investment at date t , K_t = predetermined capital stock at date t , α = capital share, σ = risk aversion coefficient and β is the subjective discount factor.

There are three shocks, namely preference shock (Z_t^c), investment specific technology shock (Z_t^x) and a TFP shock (A_t). Labour is inelastically supplied at the unit level.

The three shocks evolve as the following AR(1) processes:

$$Z_t^c - \bar{Z}^c = \rho_{zc} (Z_{t-1}^c - \bar{Z}^c) + \varepsilon_{zct},$$

$$Z_t^x - \bar{Z}^x = \rho_{zx} (Z_{t-1}^x - \bar{Z}^x) + \varepsilon_{zxt},$$

$$A_t - \bar{A} = \rho_a (A_{t-1} - \bar{A}) + \varepsilon_{at}.$$

where $\rho_{zc}, \rho_{zx}, \rho_a$ are the autoregressive coefficients of the respective shocks. The sequences $\{\varepsilon_{zct}\}$, $\{\varepsilon_{zxt}\}$, and $\{\varepsilon_{at}\}$ are mutually independent white noises associated with each of these three respective shocks and $\bar{Z}^c$, $\bar{Z}^x$, and $\bar{A}$ are the steady state levels of these three respective shocks.

- a) Derive the Euler equation carefully explaining each step and the underlying intuitions. (20 pts)
- b) Derive the steady state variables showing all your work. (20 pts)

c) Now write Dynare code for this problem fixing

$$\bar{A} = \bar{Z}^x = \bar{Z}^c = 1,$$

$$\beta = 0.99, \quad \alpha = 0.35, \quad \delta = 0.02, \quad \rho_a = 0.9, \quad \rho_{zc} = 0.9, \quad \rho_{zx} = 0.9, \quad \sigma = 1.5.$$

Make sure to compute the implicit real wage and rental prices of capital. (30 pts)

d) Based on your Dynare output, write a report within 1000 words explaining the key results. Your report should include explanations of the variance decompositions of shocks and the impulse responses to all shocks. (30 pts)

[You must submit your code and Dynare output together with your report]

Overall word limit: 1000 words